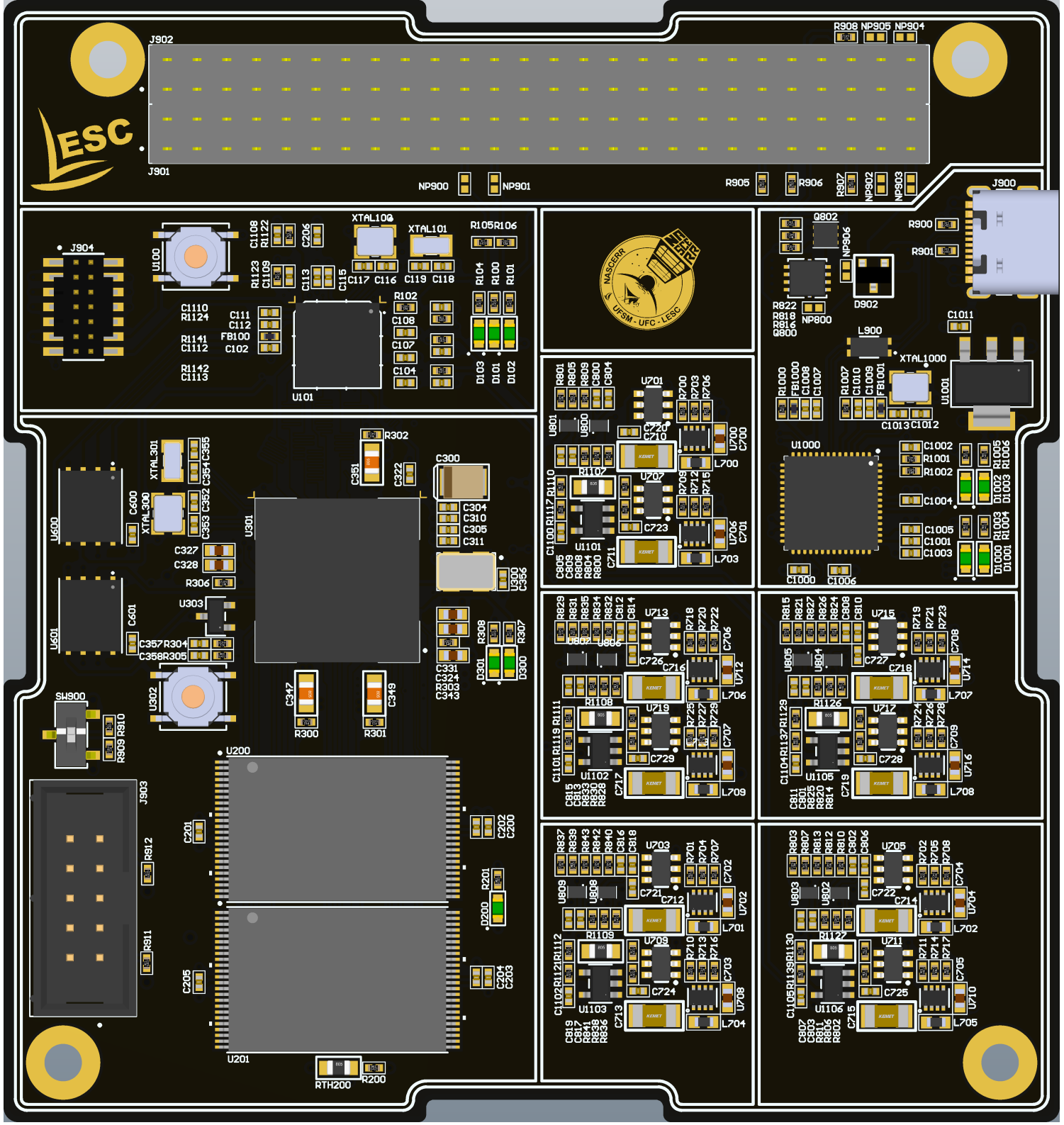
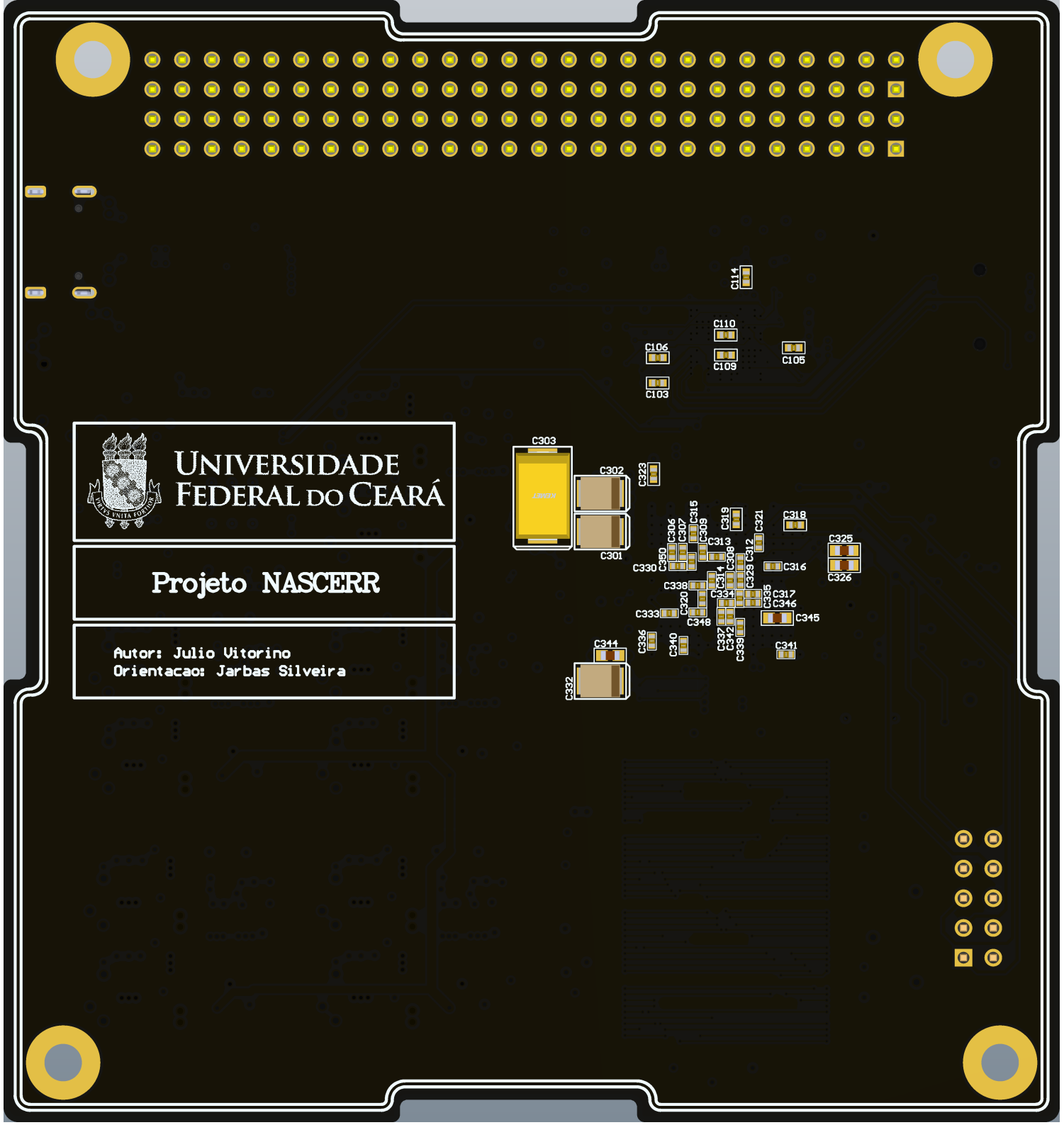
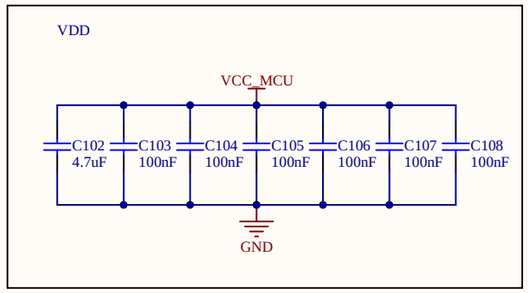
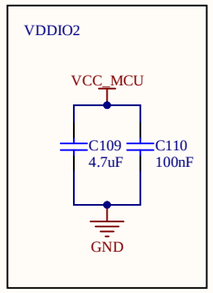
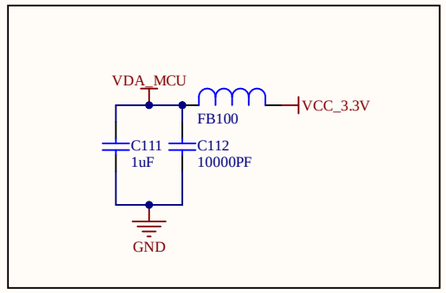
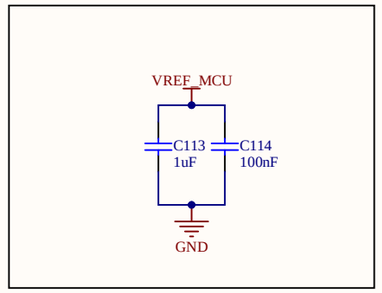

LESC
NASCERR
UFSM - UFC - LESC
J902
J901
J900
J904
J903
SW900
U100
U101
U200
U201
U300
U301
U302
U303
U600
U601
U700
U701
U702
U703
U705
U706
U708
U709
U712
U713
U714
U715
U716
U717
U801
U802
U803
U804
U805
U809
U1000
U1001
U1101
U1102
U1103
U1104
U1105
U1106
XTAL100
XTAL101
XTAL300
XTAL301
XTAL1000
Q802
D902
L900
L700
L701
L702
L703
L704
L705
L706
L707
L708
L709
SH900
RTH200

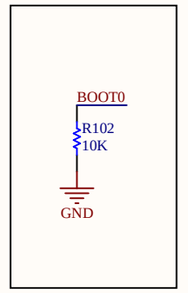

UNIVERSIDADE FEDERAL DO CEARÁ
Projeto NASCERR
Autor: Julio Vitorino
Orientacao: Jarbas Silveira
C114
C110
C106
C109
C105
C103
C303
C302
C323
C301
C315
C319
C321
C318
C306
C307
C309
C313
C312
C325
C350
C330
C308
C329
C316
C326
C314
C338
C331
C335
C317
C320
C346
C333
C348
C345
C336
C340
C337
C342
C339
C341
C344
C332

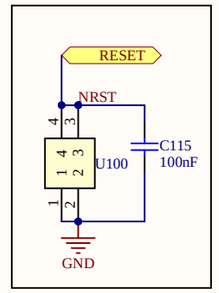


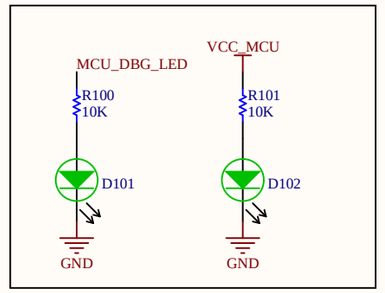


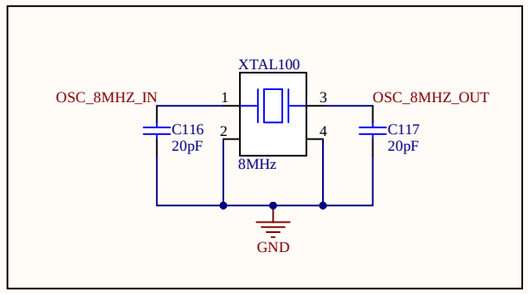


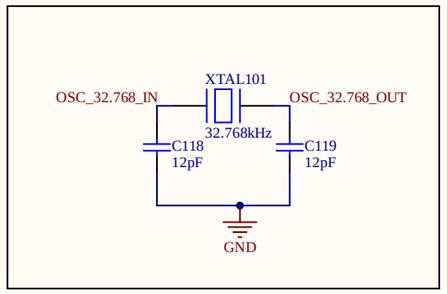


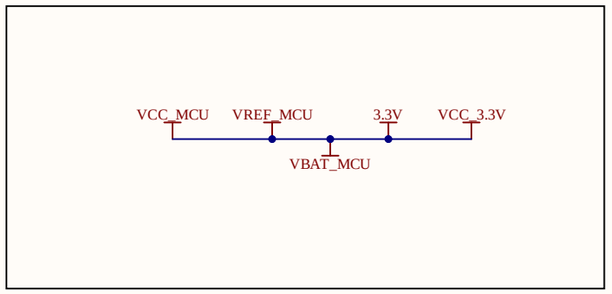


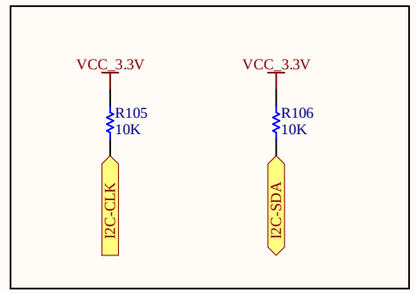


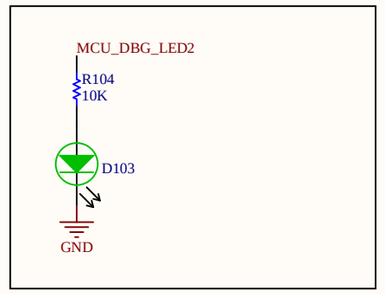




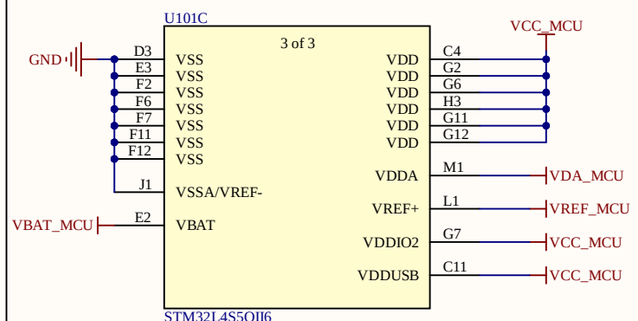


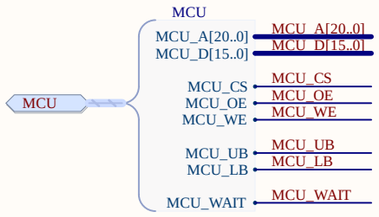


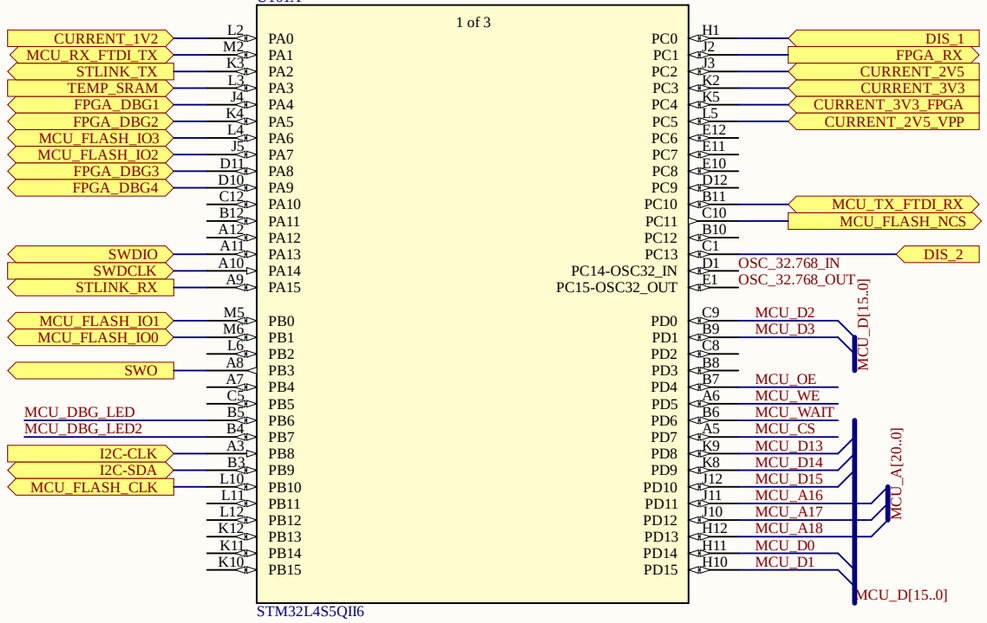


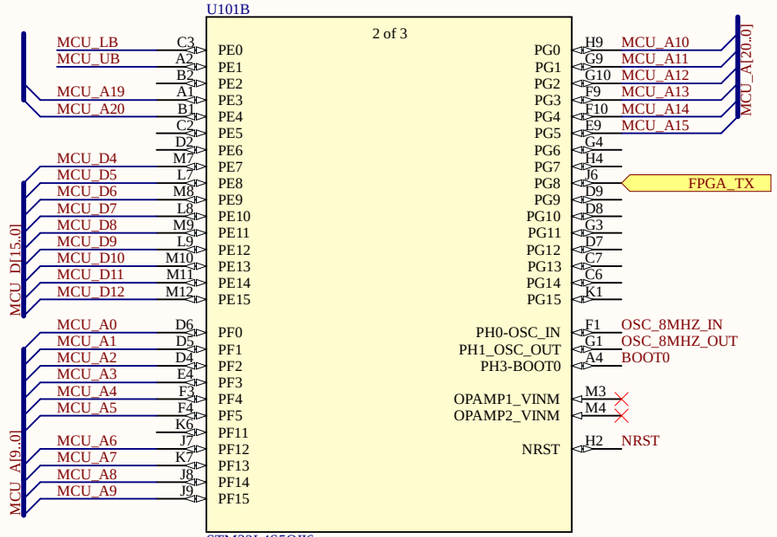


MCU



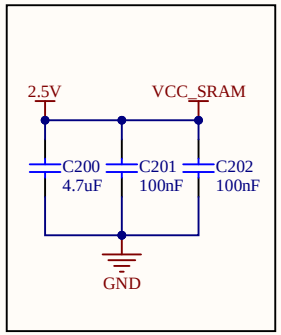


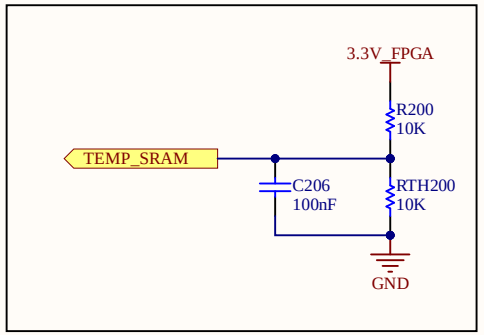


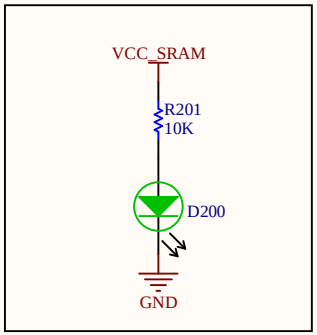


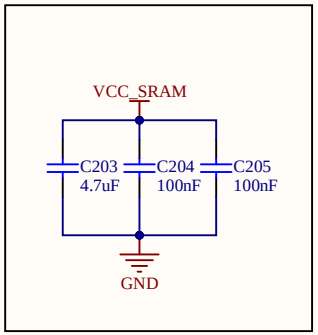
Nome da Sheet: "

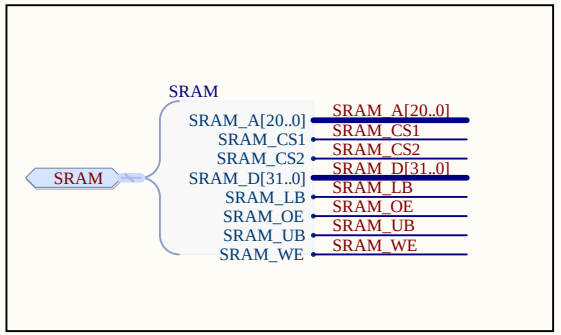
LESC - Laboratório de Engenharia e Sistemas de Computação	
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970	
Data:	
Projetista: Julio Vitorino	
PCB Design: Julio Vitorino	Revision:
	Size:



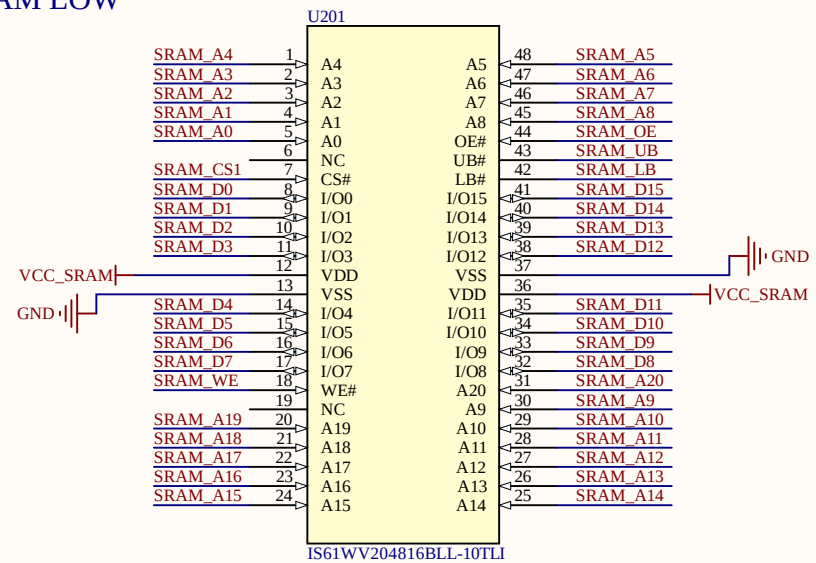




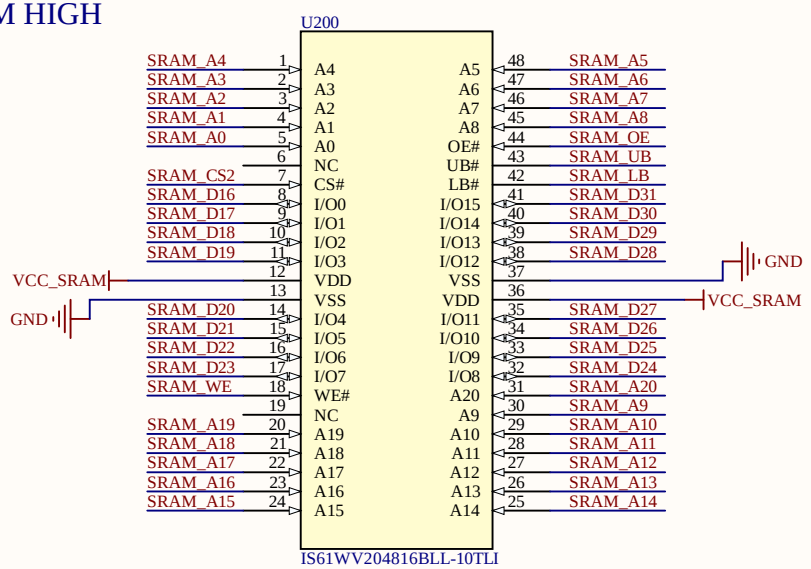




SRAM LOW



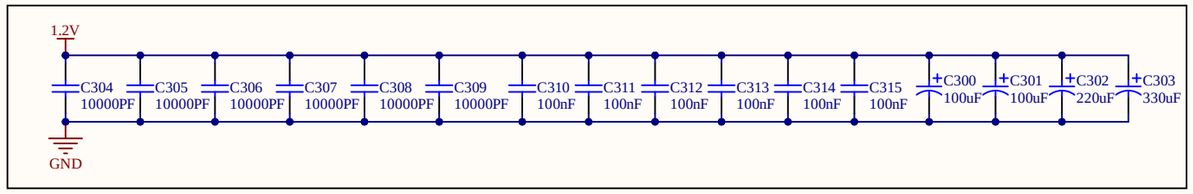
SRAM HIGH

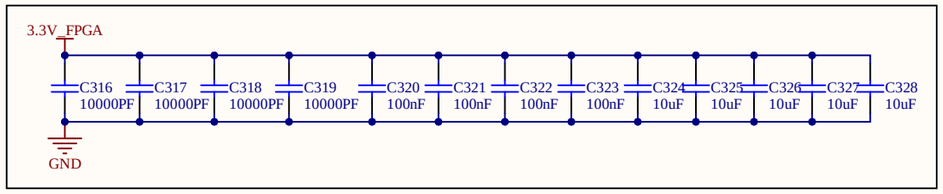


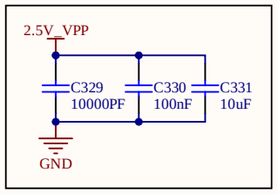
Nome da Sheet: "

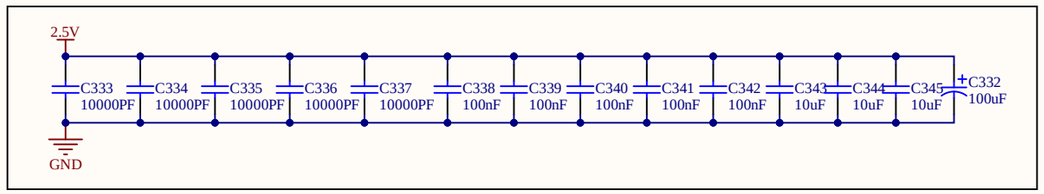
LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

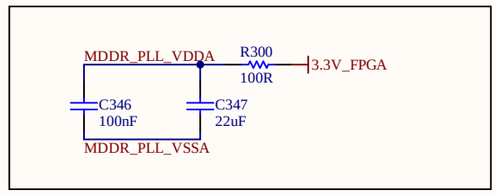
Data:	
Projetista: Julio Vitorino	
PCB Design: Julio Vitorino	Revision:
	Size: A4

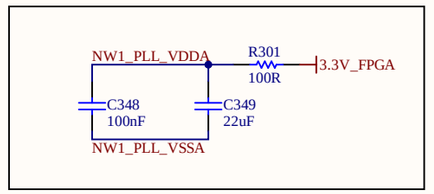


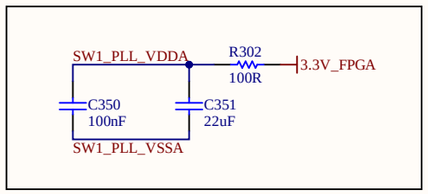


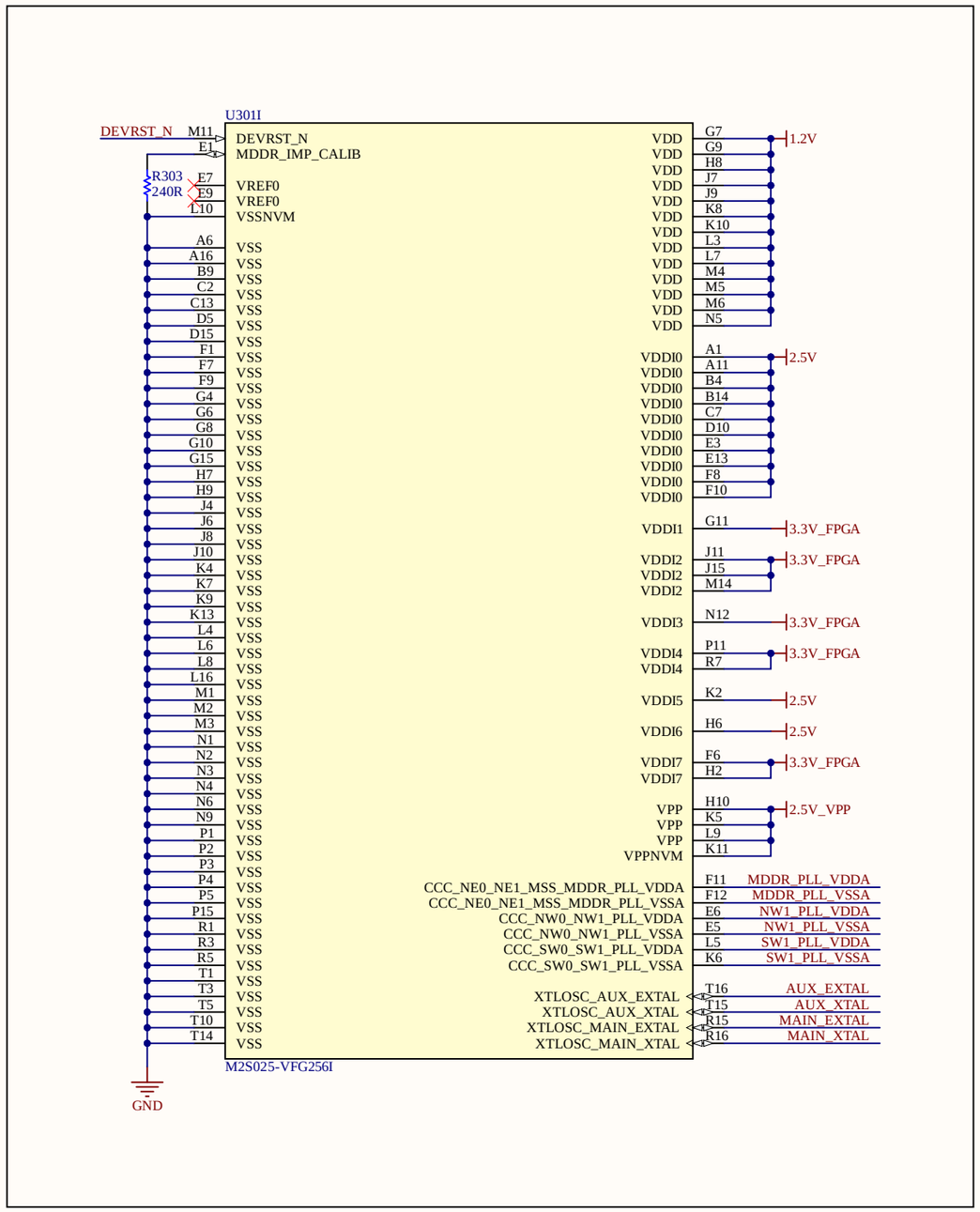


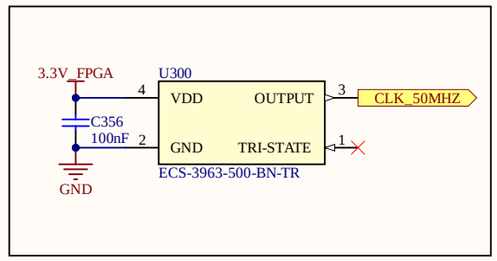


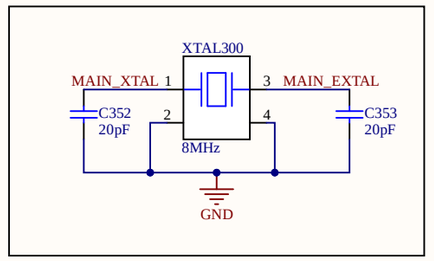


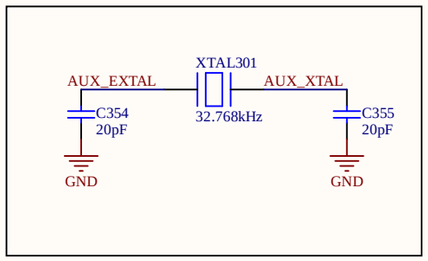


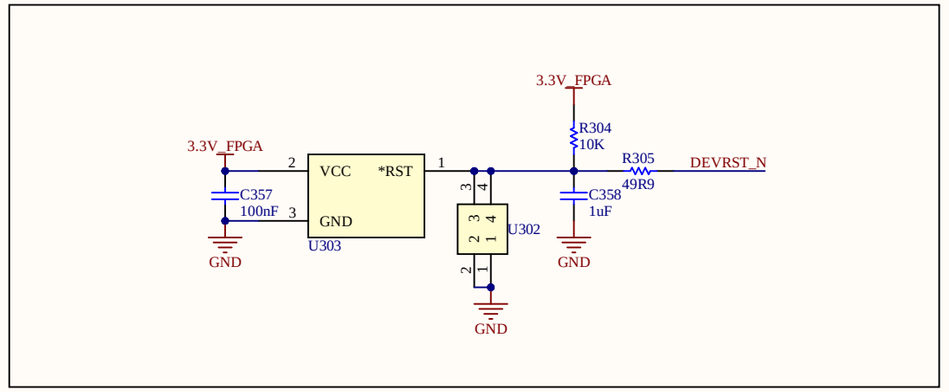


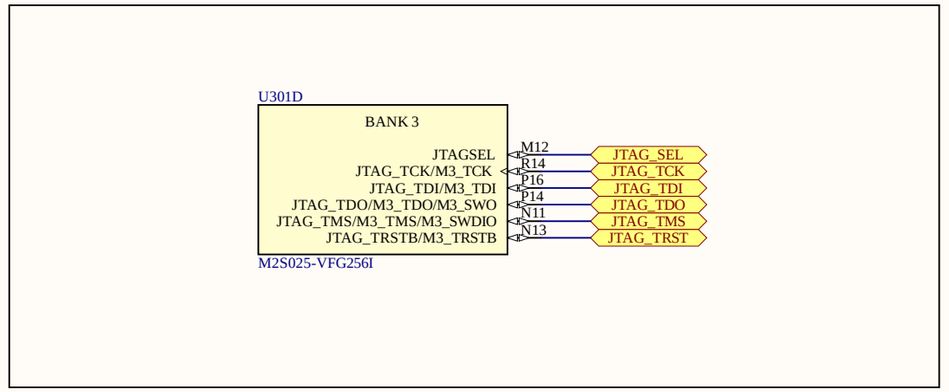








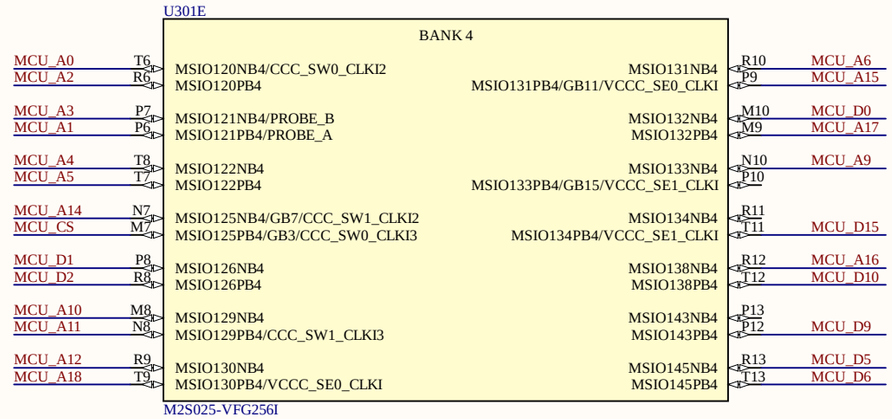


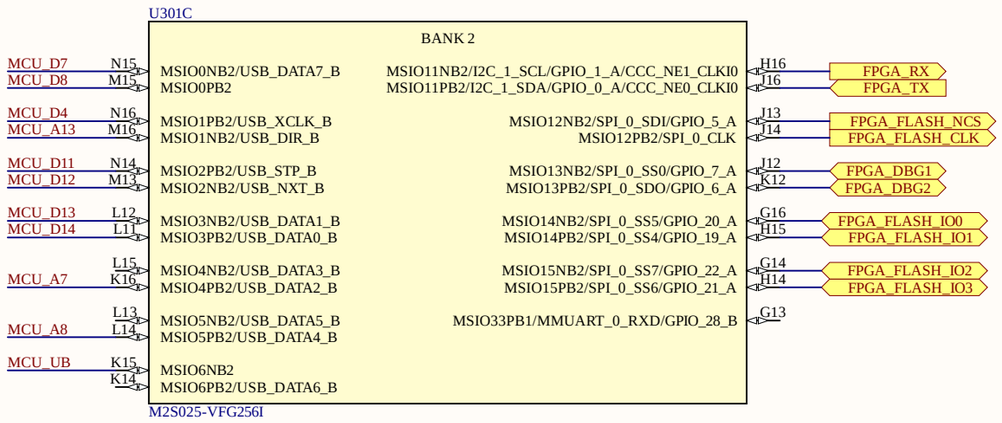


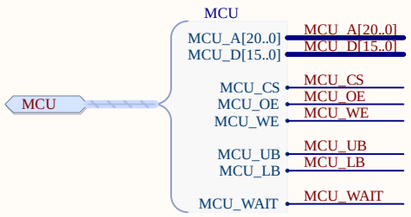
Nome da Sheet: "

LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

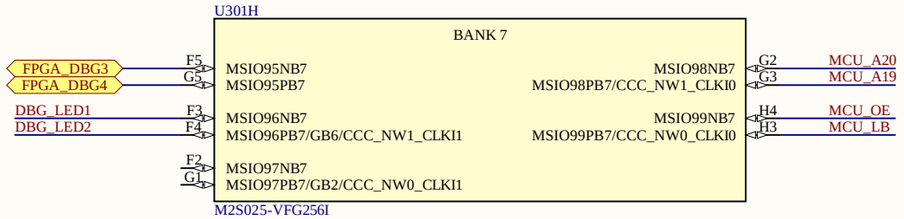
Data:	
Projetista: Julio Vitorino	
PCB Design: Julio Vitorino	Revision:
	Size:



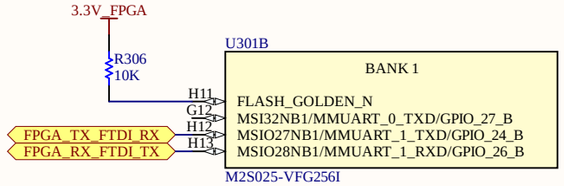




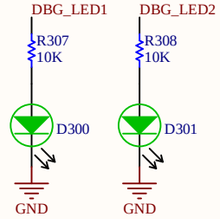
FMC INTERFACE

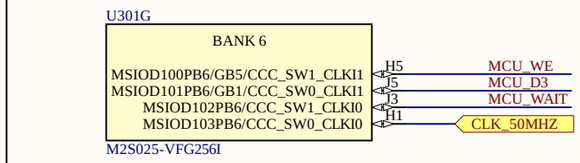


FLASH INTERFACE



FTDI DEBUG INTERFACE





Nome da Sheet: "

LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

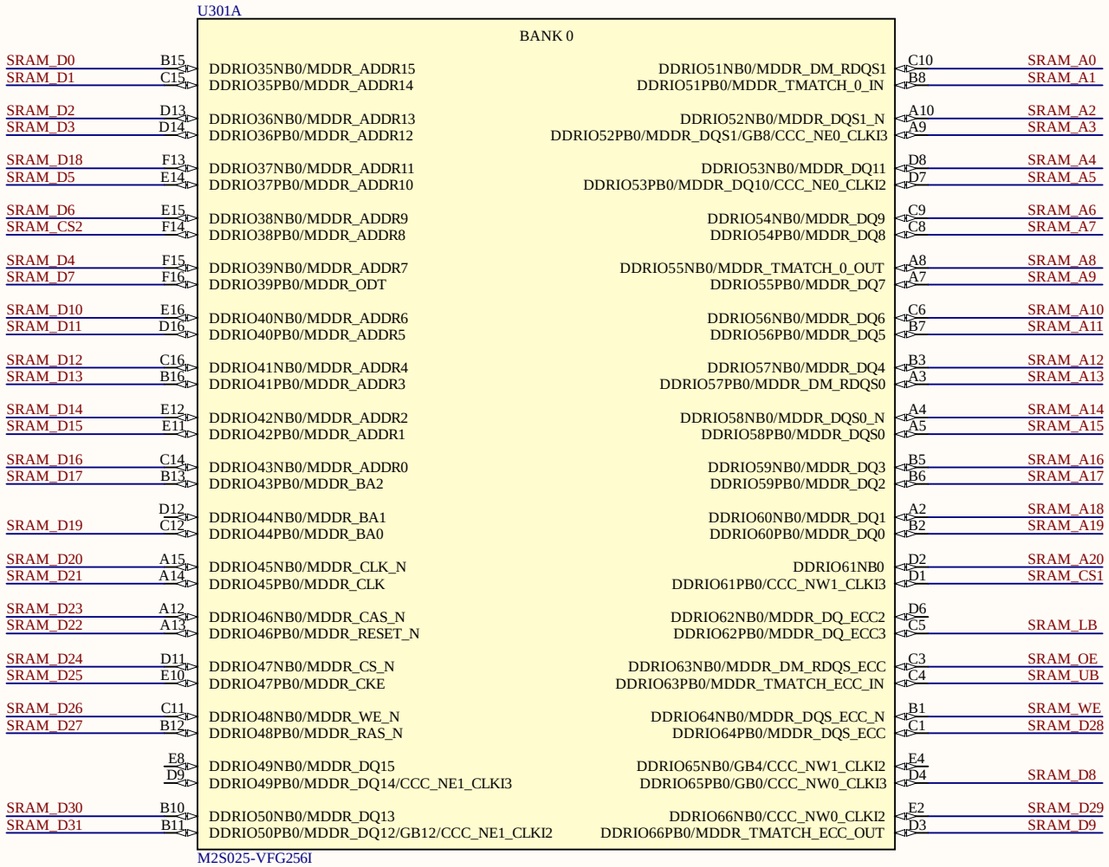
Data:

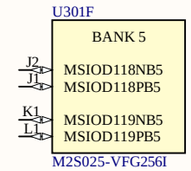
Projetista: Julio Vitorino

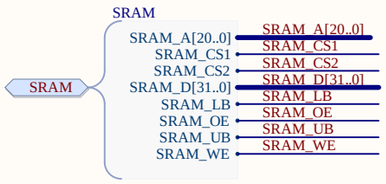
PCB Design: Julio Vitorino

Revision:

Size:







Nome da Sheet: "

LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

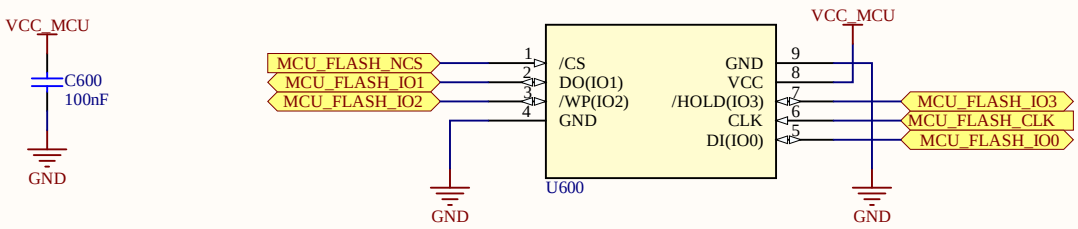
Data:

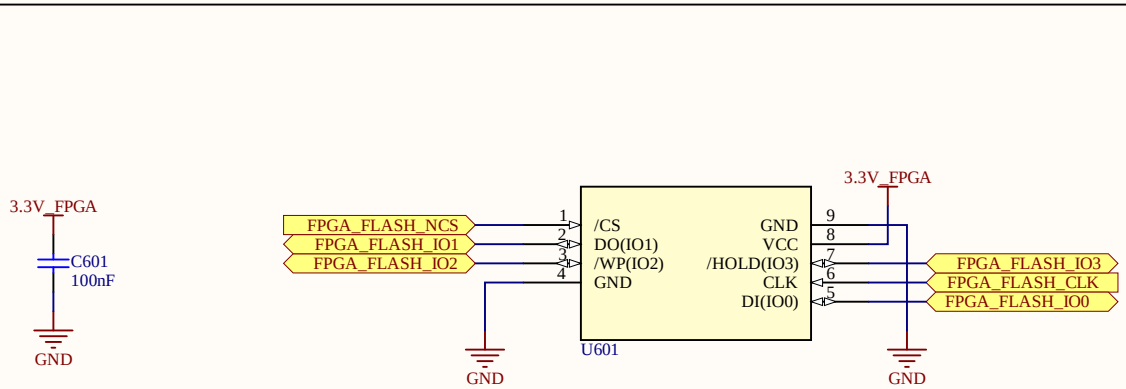
Projetista: Julio Vitorino

PCB Design: Julio Vitorino

Revision:

Size:





Nome da Sheet: 0 - TOP

LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

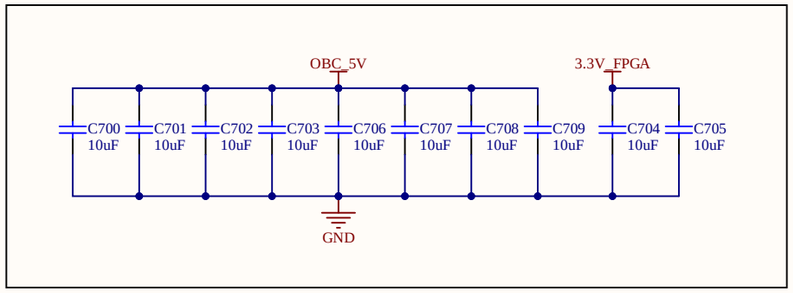
Data: 25/11/2021

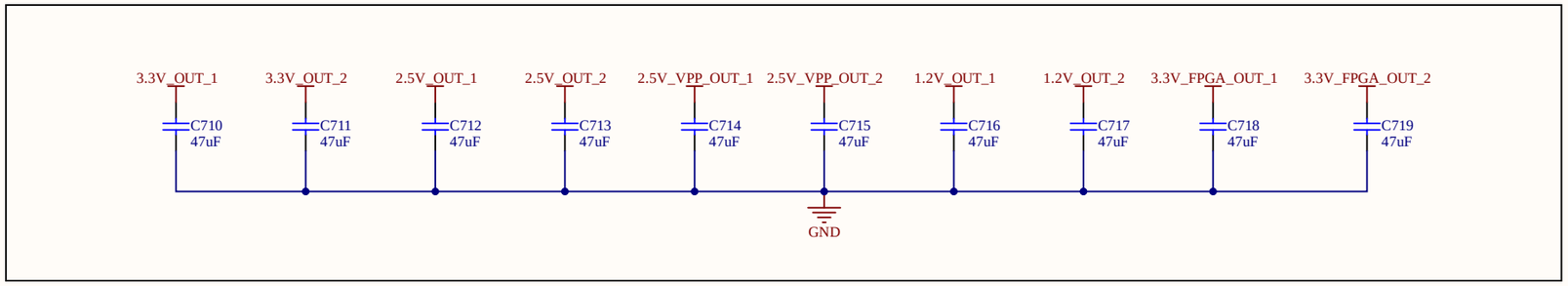
Projetista: Julio Vitorino

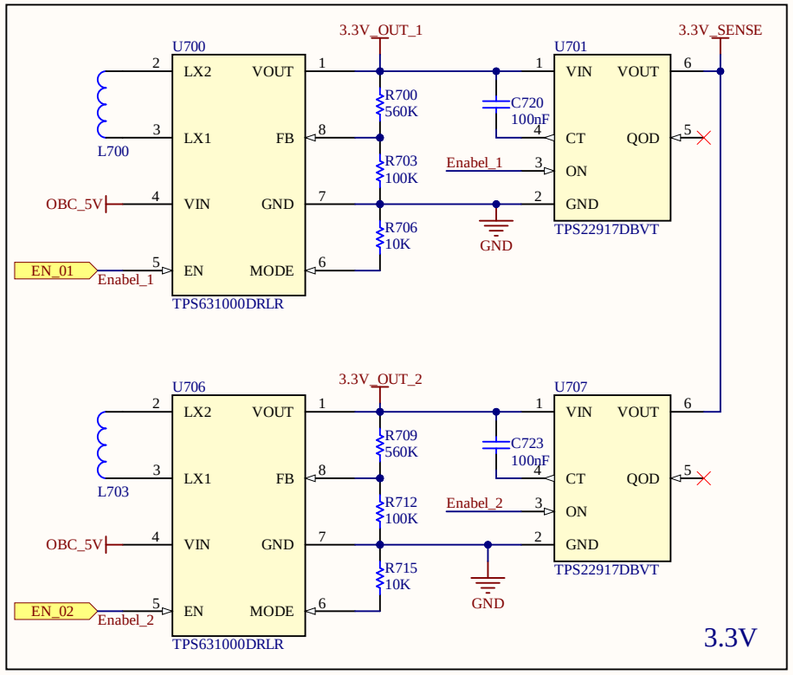
PCB Design: Julio Vitorino

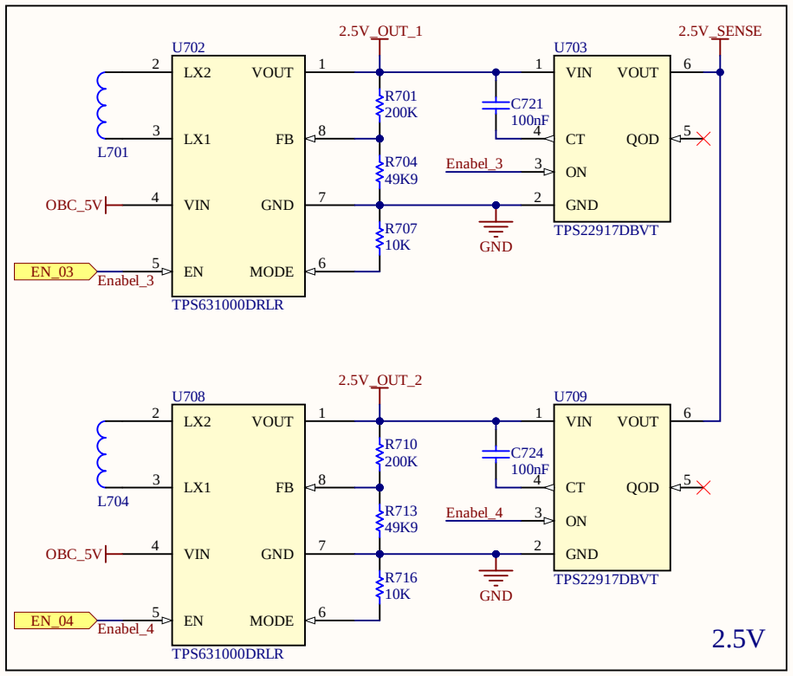
Revision: 01

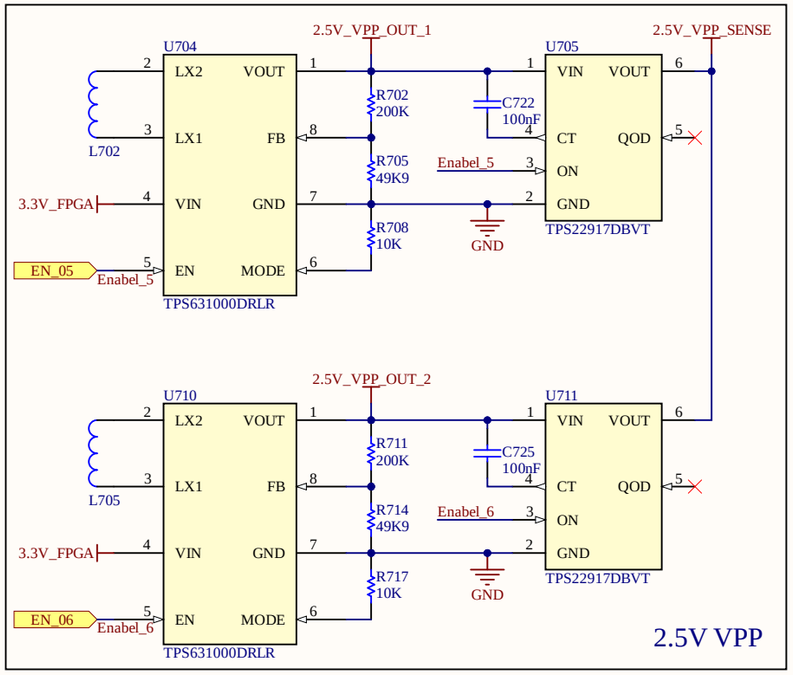
Size: A4

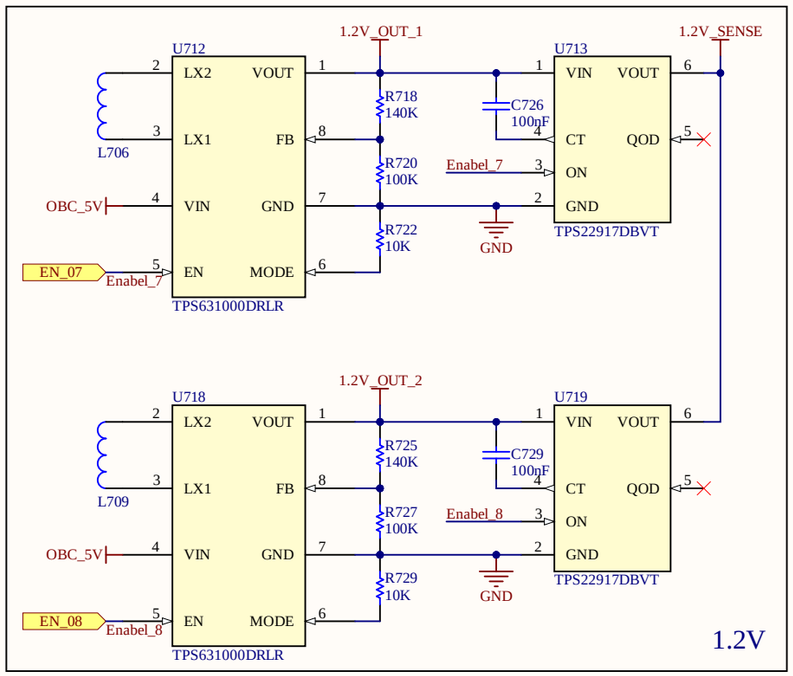


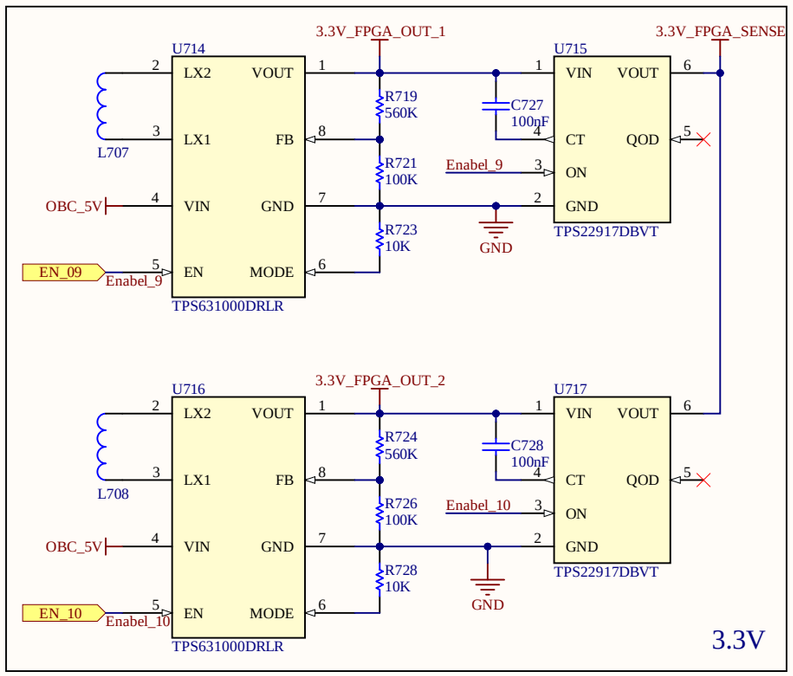






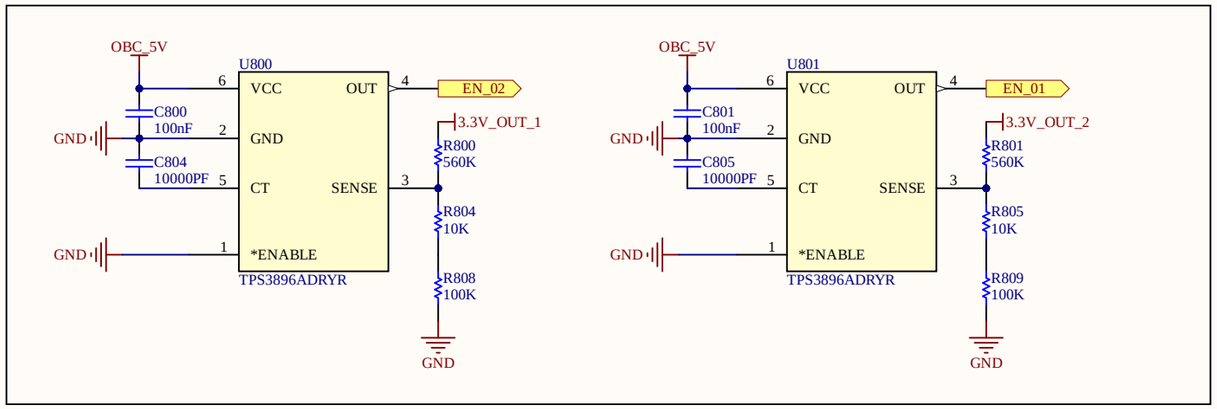


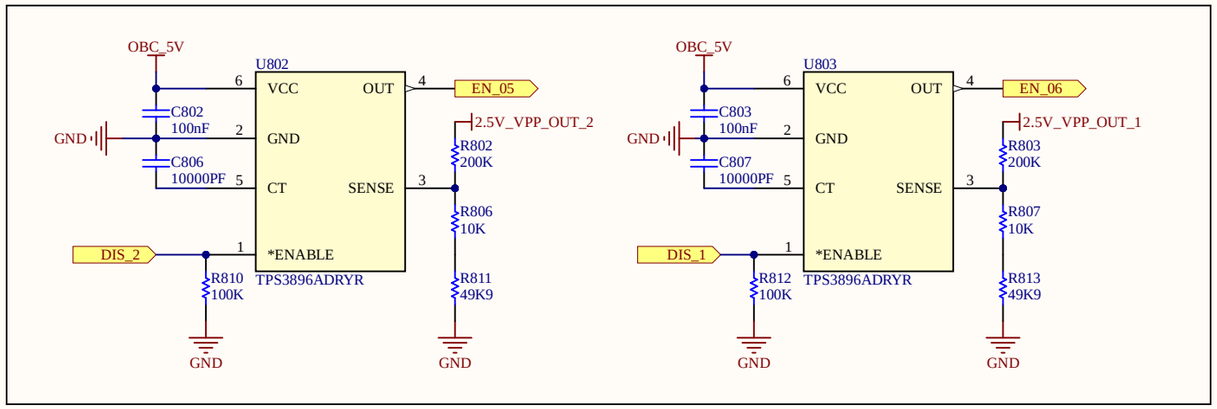


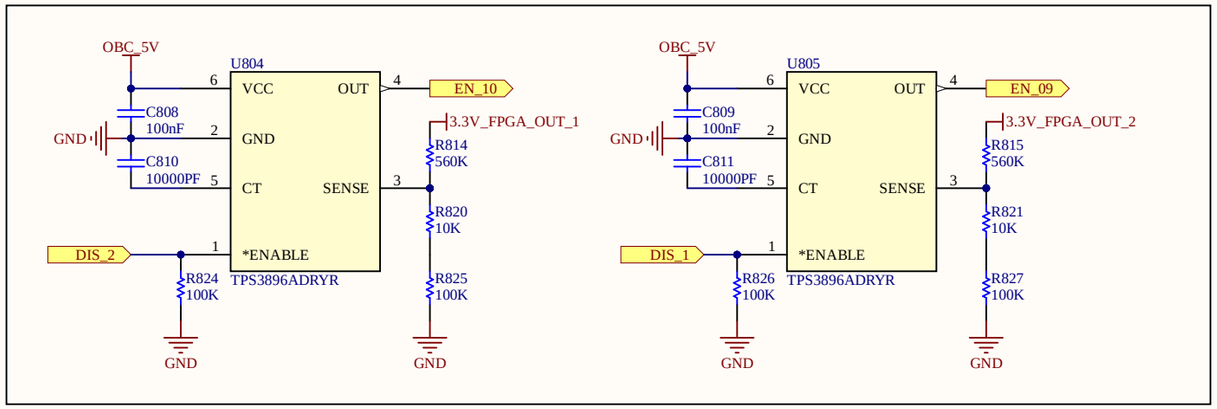


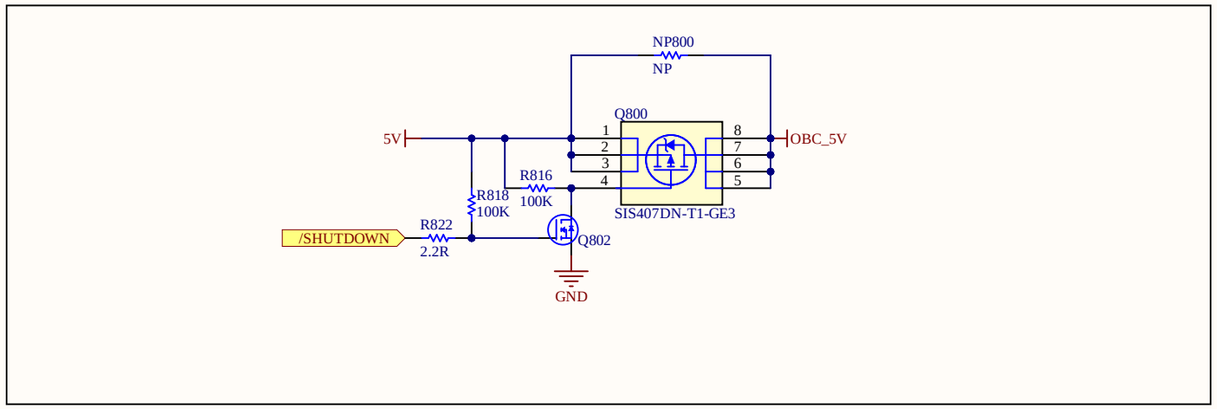
Nome da Sheet: 0 - TOP

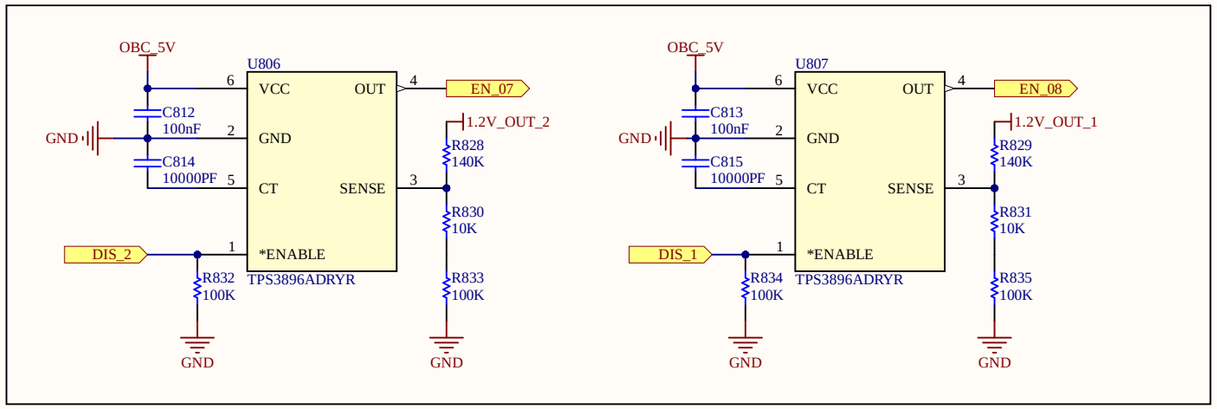
LESC - Laboratório de Engenharia e Sistemas de Computação	
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970	
Data: 25/11/2021	
Projetista: Julio Vitorino	
PCB Design: Julio Vitorino	Revision: 01
	Size: A4

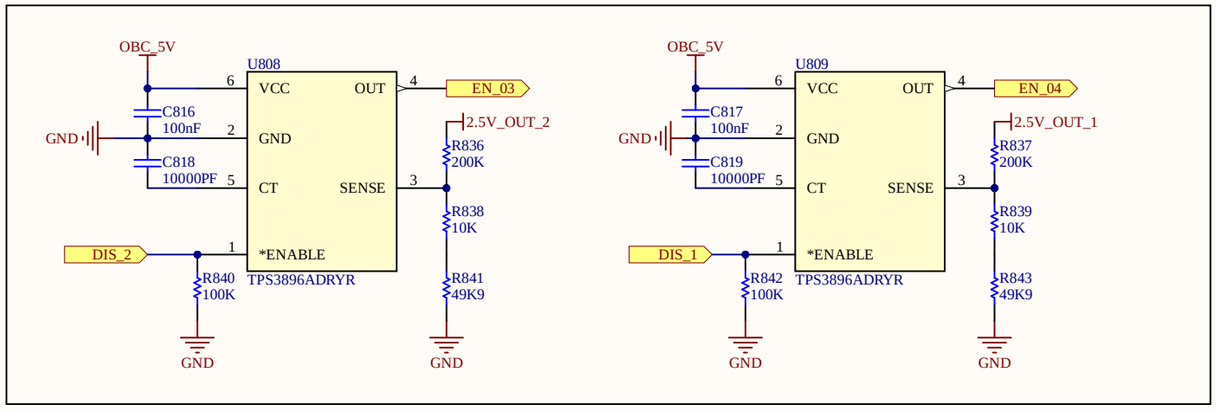










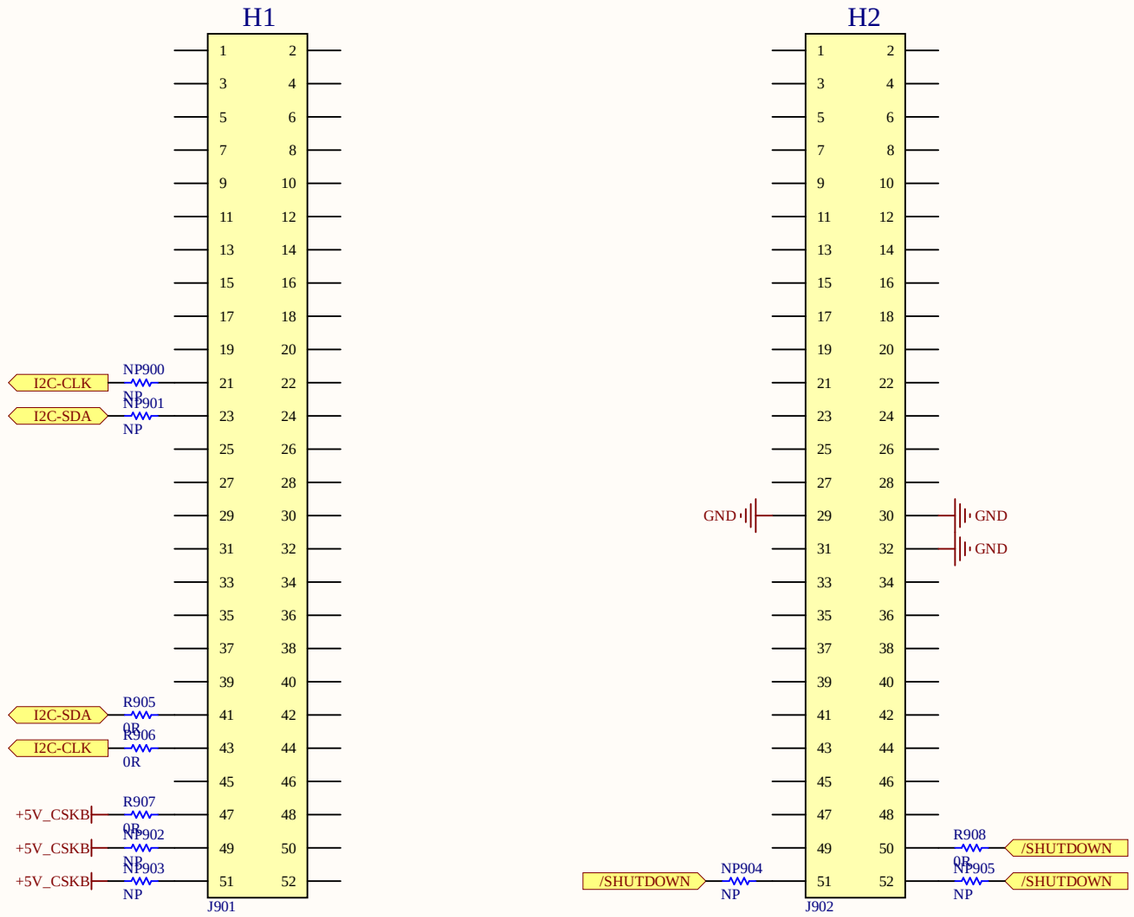


Nome da Sheet: 0 - TOP

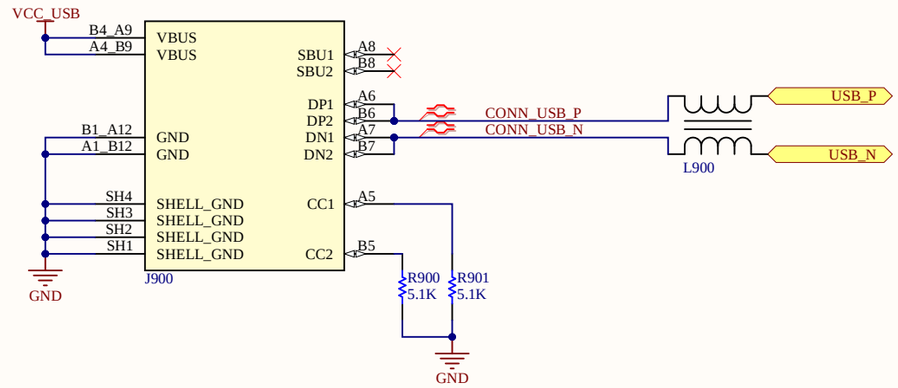
LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

Data: 25/11/2021
Projetista: Julio Vitorino
PCB Design: Julio Vitorino
Revision: 01
Size: A4

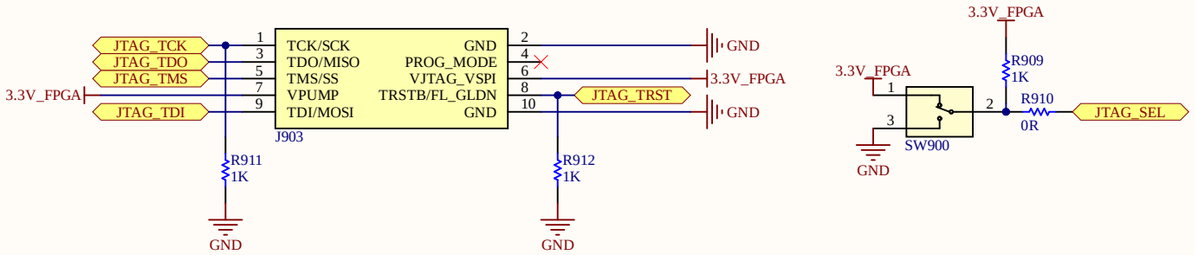
CUBESAT BUS



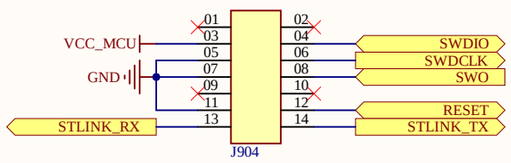
USB DEBUG VCP

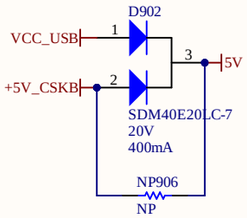


FPGA PROGRAM/DEBUG



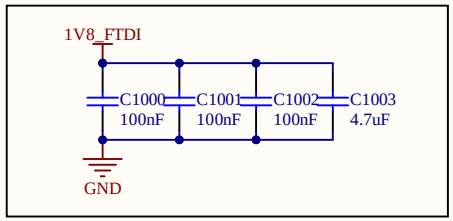
MCU PROGRAM/DEBUG

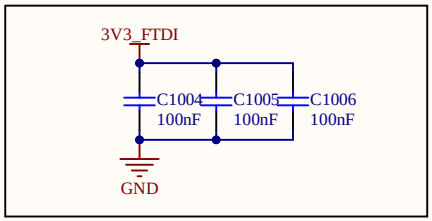


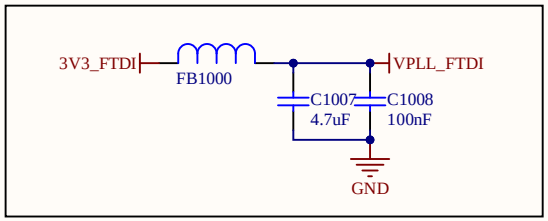


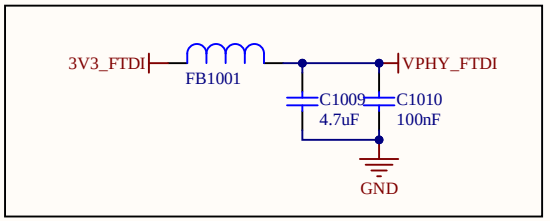
Nome da Sheet: "

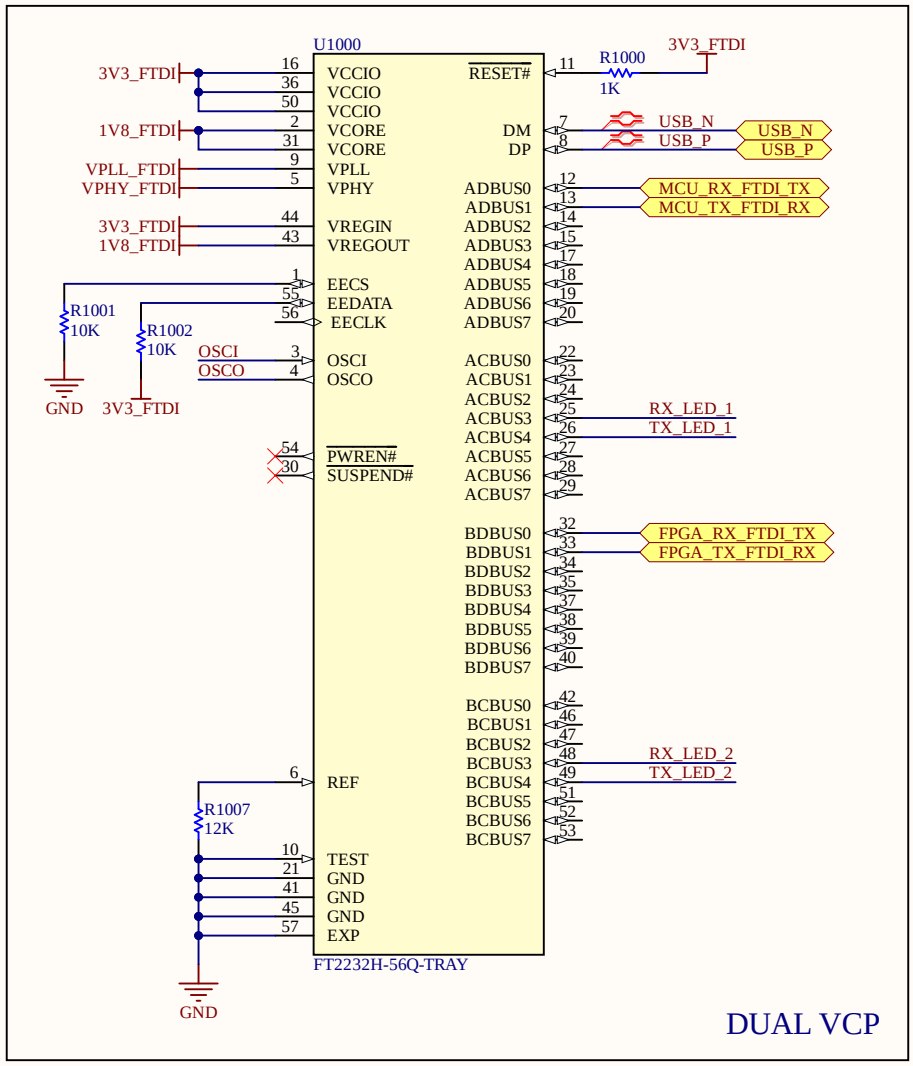
LESC - Laboratório de Engenharia e Sistemas de Computação	
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970	
Data:	
Projetista: Julio Vitorino	
PCB Design: Julio Vitorino	Revision:
	Size:

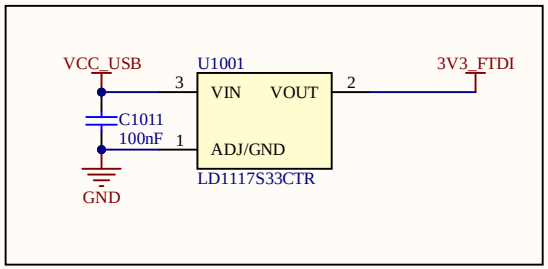


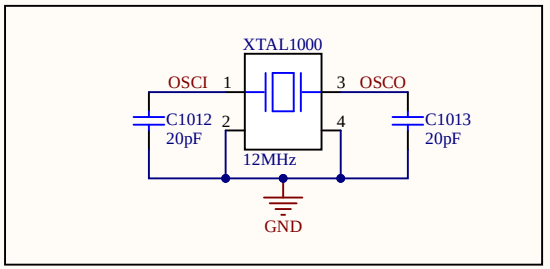


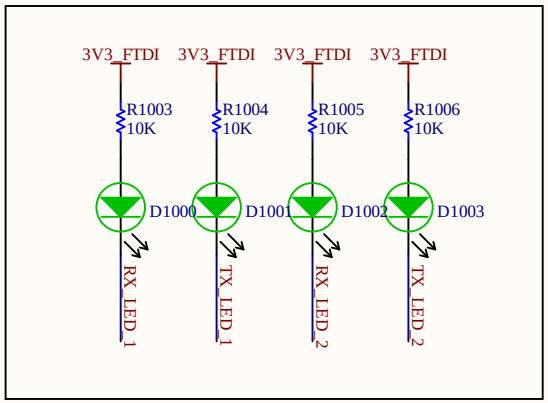








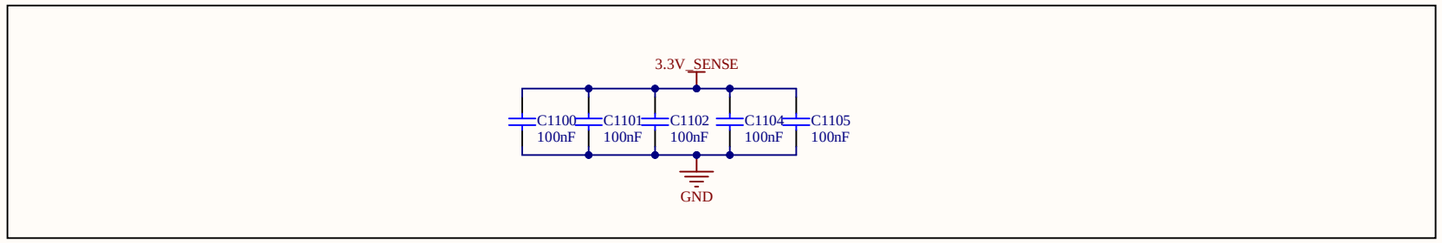


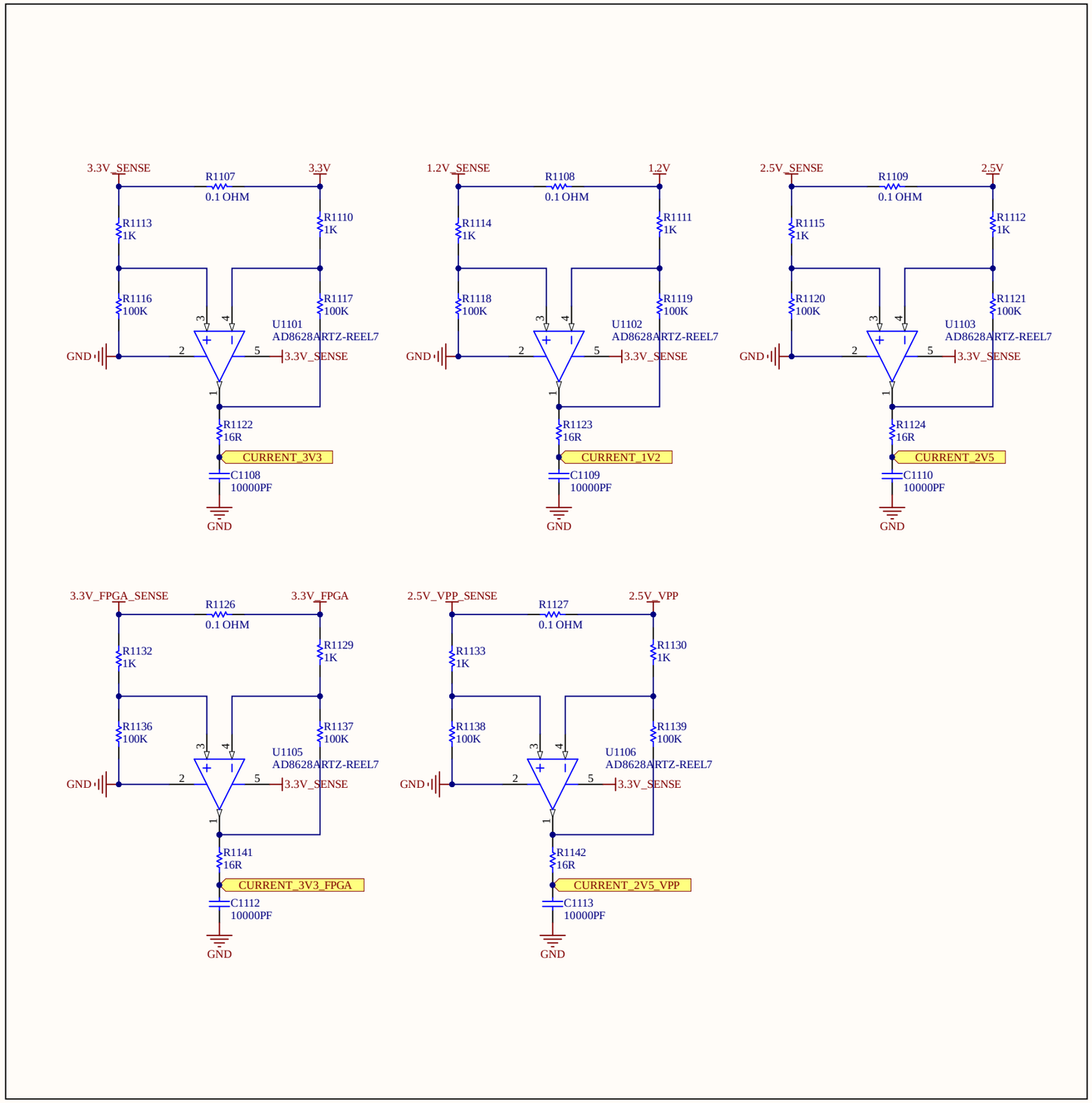


Nome da Sheet: "

LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

Data:	
Projetista: Julio Vitorino	
PCB Design: Julio Vitorino	Revision:
	Size: A4





Nome da Sheet: "

LESC - Laboratório de Engenharia e Sistemas de Computação
Campus do Pici - Bloco 723, S/N - Pici, Fortaleza - CE, 60455-970

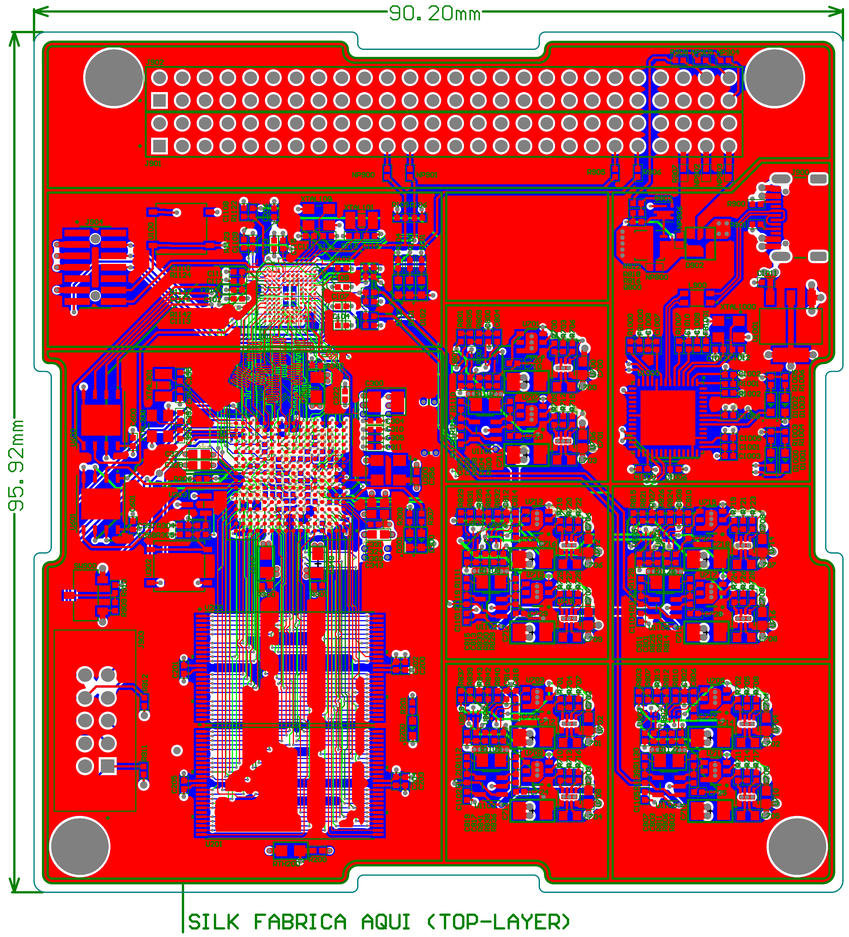
Data:

Projetista: Julio Vitorino

PCB Design: Julio Vitorino

Revision:

Size:



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca
☐ HASL–Hot Air Solder Leveling		☐ Preta
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca
☐ Ouro Quimico		☐ Preta
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

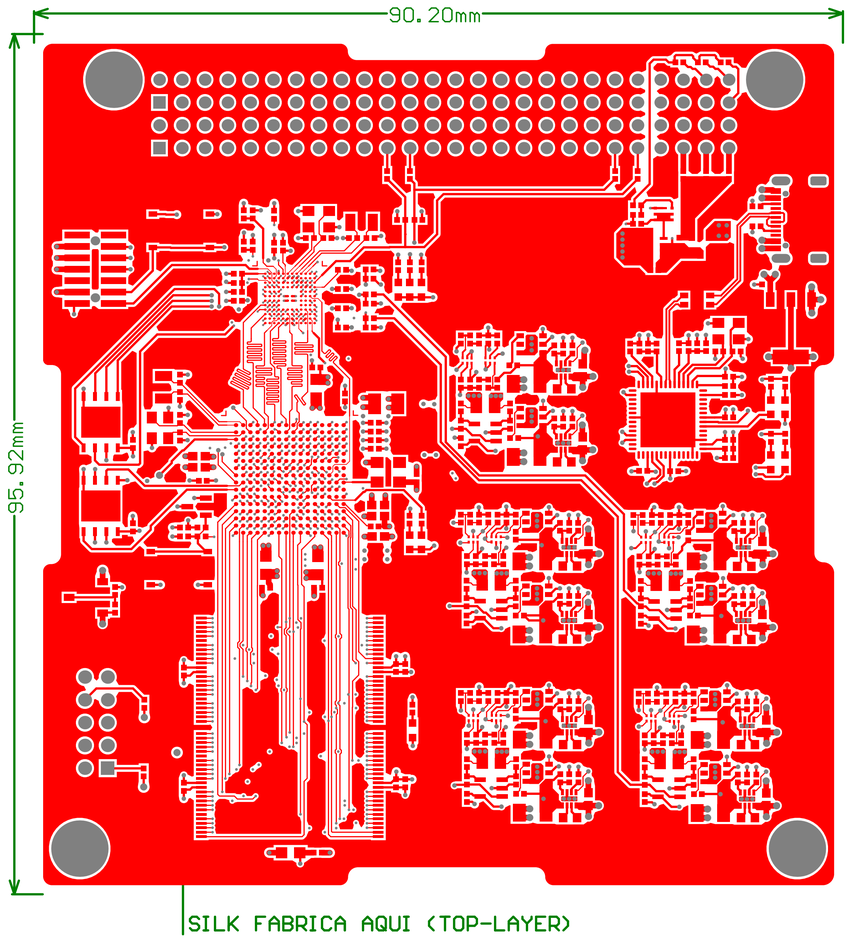
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca ☐ Preta
☐ Ouro Quimico		
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

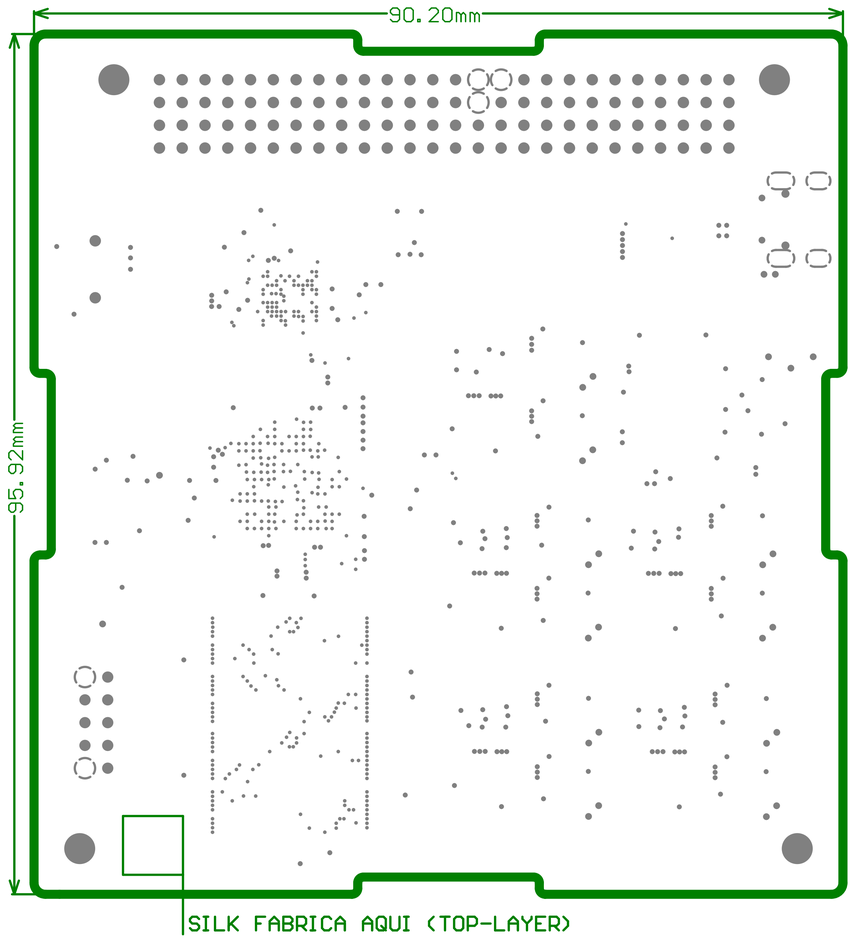
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade		
☐ Ouro Quimico	■ Lado Solda	■ Branca ☐ Preta
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

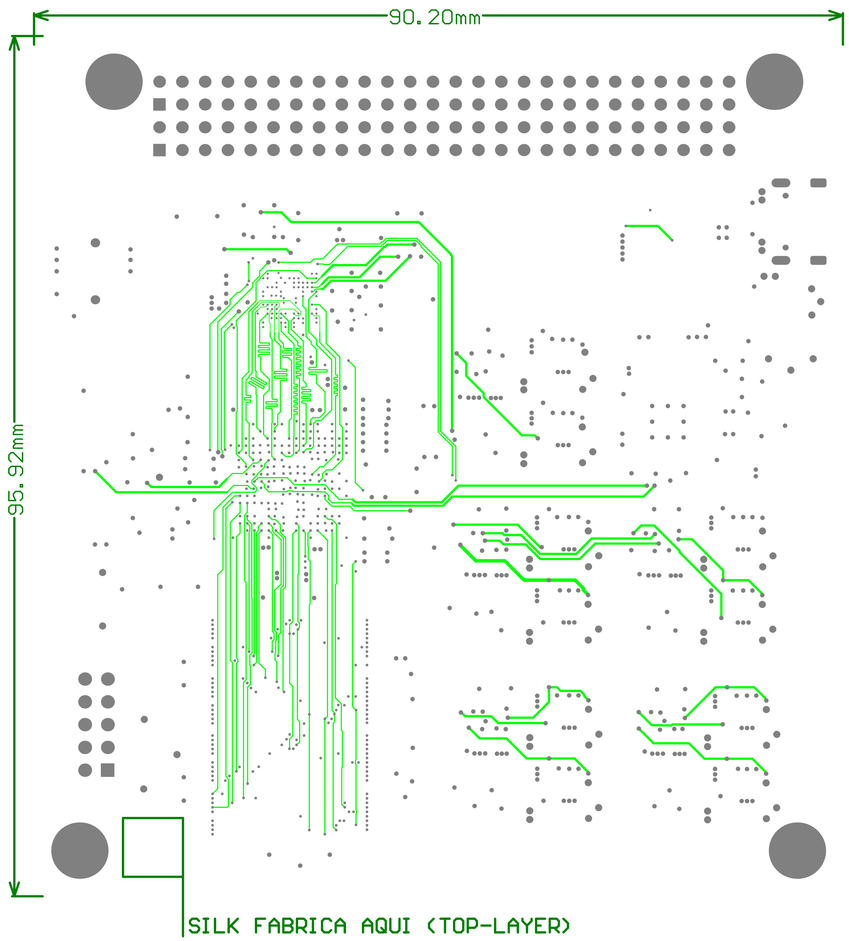
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca ☐ Preta
☐ Ouro Quimico		
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

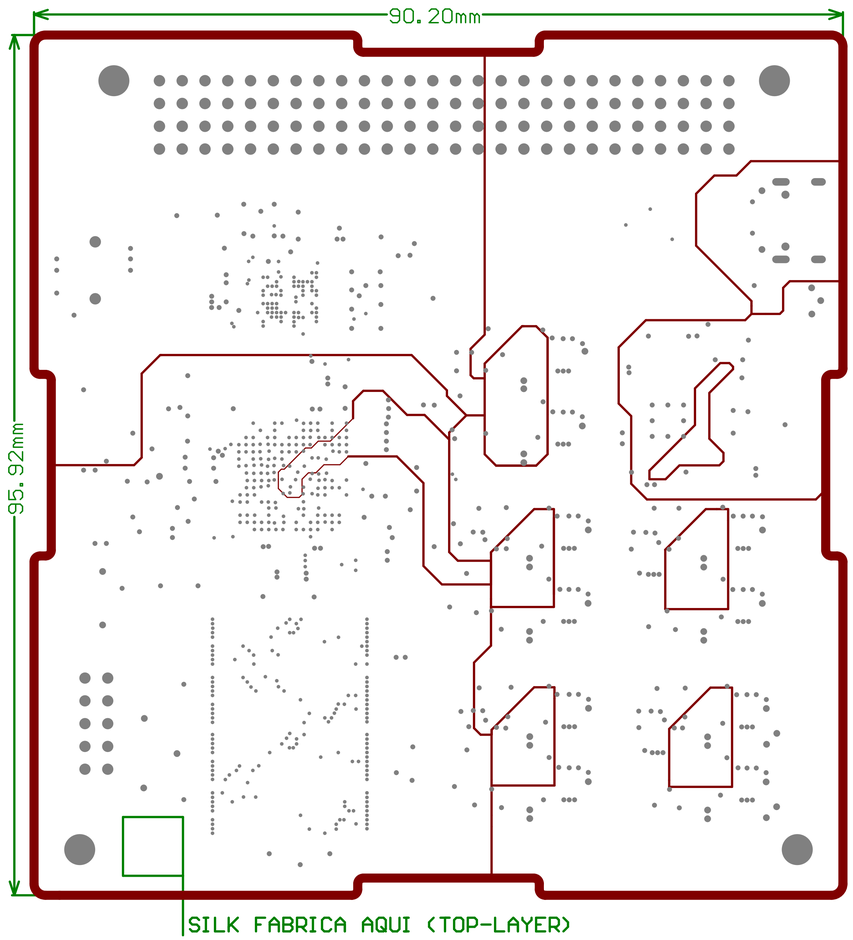
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca ☐ Preta
☐ Ouro Quimico		
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

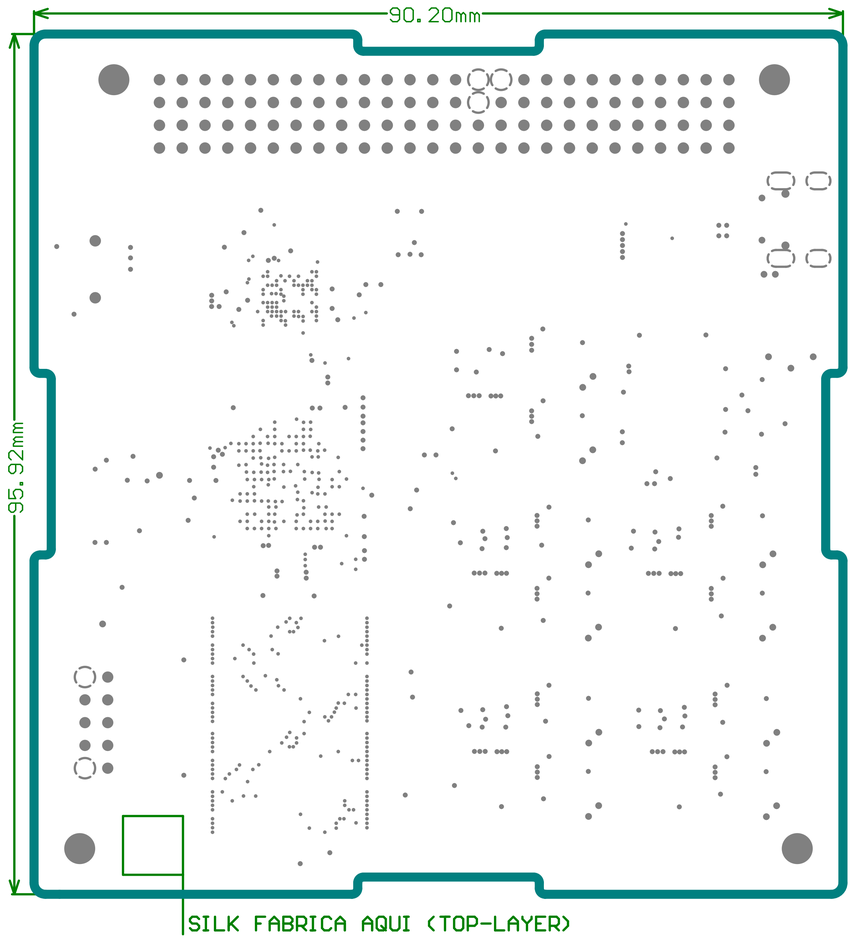
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade		
☐ Ouro Quimico	■ Lado Solda	■ Branca ☐ Preta
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

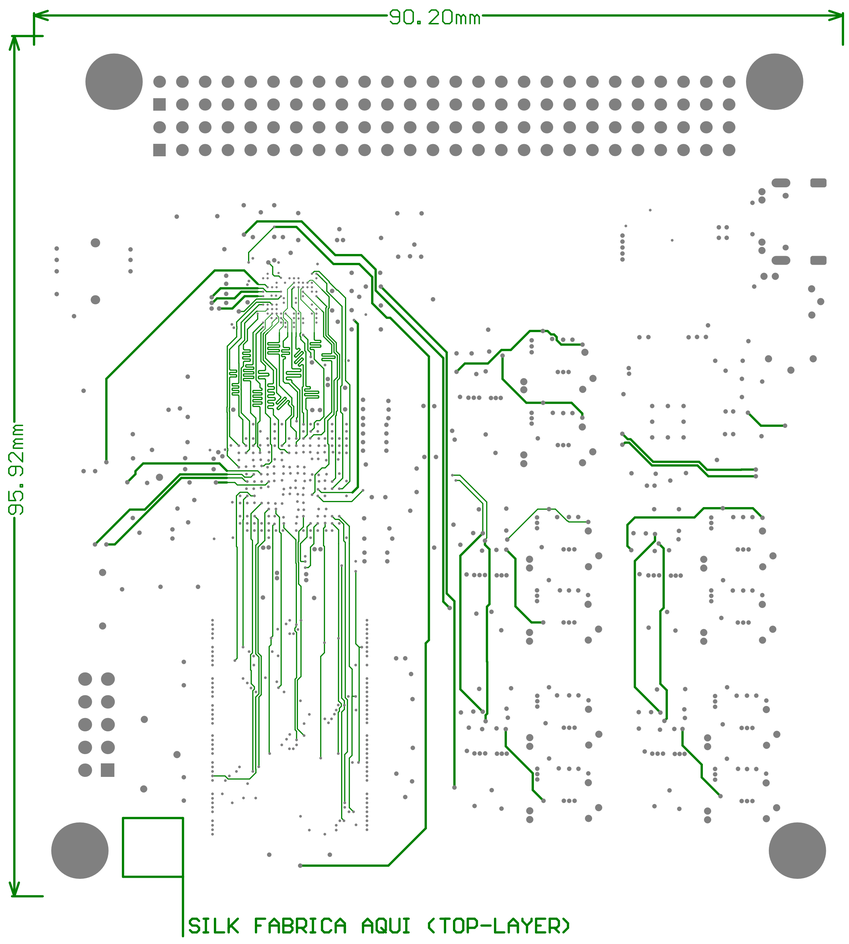
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca ☐ Preta
☐ Ouro Quimico		
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

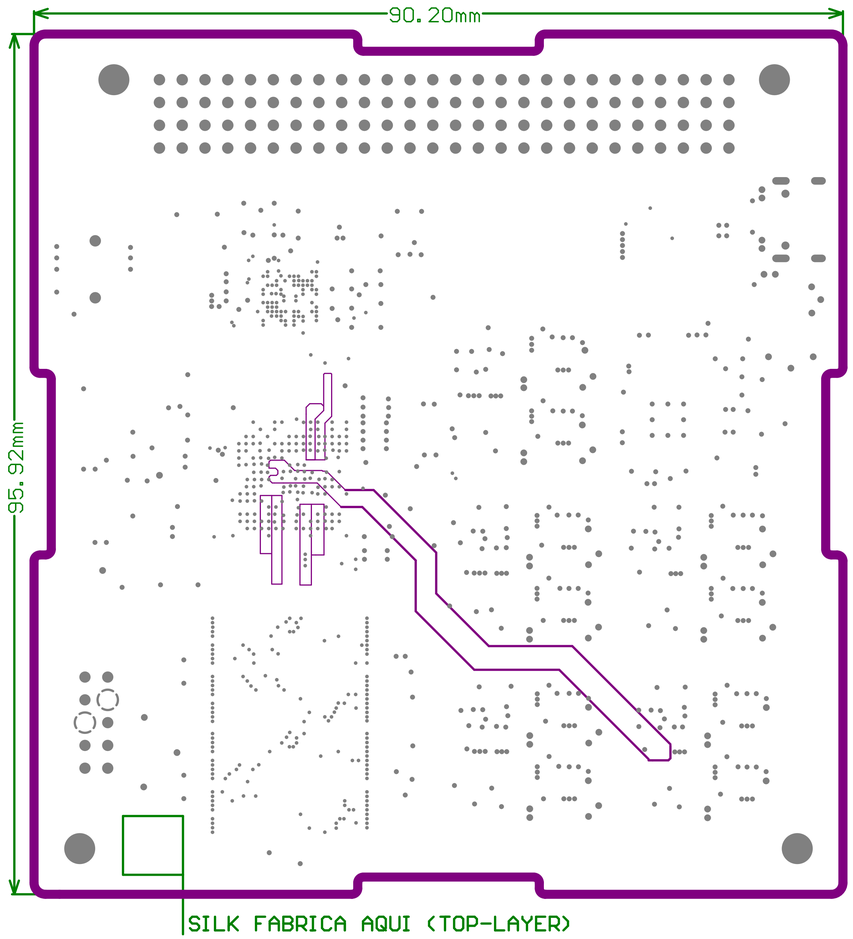
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca ☐ Preta
☐ Ouro Quimico		
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

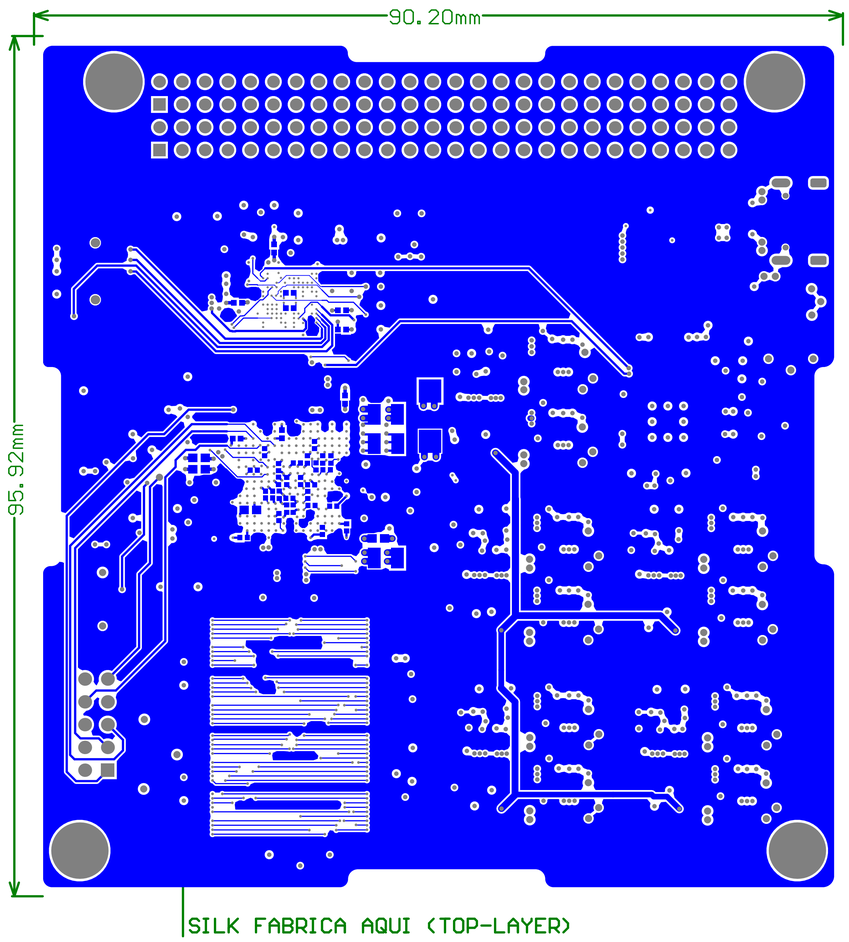
Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz



Especificacao Para Fabricacao

Identificacao da placa

Descricao: PCI NASCERR RV01		
ID: NOVO	Revisao: 01A	Data: 09.09.2024

Modelo solicitado

Material Base	Esp. Mat.	Cam. de Cobre	Num. Face
■ FR4	☐ 0,8mm	☐ 1/4 Oz – 8.5um	☐ 01
☐ FR2	☐ 1,0mm	■ 1/2 Oz – 17um	☐ 02
☐ CEM1	☐ 1,2mm	☐ 1 Oz – 35um	☐ 04
☐ CEM2	■ 1,6mm	☐ 2 Oz – 70um	☐ 06
☐	☐	☐	■ 08

Acabamento Superficie	Serigrafia	
	Aplicacao	Cor
☐ Verniz antioxidante	■ Lado Componente	■ Branca ☐ Preta
☐ HASL–Hot Air Solder Leveling		
☐ OSP–Prot. Org. Soldabilidade	■ Lado Solda	■ Branca ☐ Preta
☐ Ouro Quimico		
■ Ouro Eletrolitico (ENIG)		

Blank	Teste Eletrico
☐ Sim /Qtd:	■ Sim
■ Nao	☐ Nao

Acabamento Mecanico	Mascara anti–solda	
	Aplicacao	Cor
☐ Fresa CNC	☐ UV	☐ Verde
☐ Vinco (Detalhe B)	■ Photo Imageable	■ Preta
☐ Manual	☐ SnPb	☐ Azul
☐ Outro:		

Tolerancias:
Mecanica: +/– 0,2mm
Furo: +/– 0,05mm

Area util

90,20mm X 95,92mm

Observacoes Gerais

Especura cobre interno: 1 Oz
Especura cobre externo: 1 Oz